



B.Sc. Computer Science

Division	Faculty of Engineering and Technology
School Name	School of Computer Science & Engineering
Department Name	Department of Computer Engineering and Technology
Programme Name	B.Sc. Computer Science

Semester	Odd (I)	Even (II)	Total Credits
First Year	22	21	43
Second Year	22	22	44
Third Year	21	22	43
Fourth Year	16	19	35

+ + + + + + + + + + + **COURSE BASKET** + + + + + + + + + + +

| Course Type | Description |
|----------------------|--|
| Programme Core | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU |
| Programme Electives | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students. |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core. |

Semester I

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|---------------------------------------|---------------|
| I | UC | Indian Constitution | 1 |
| I | UC | Environment and Sustainability | 1 |
| I | UC | Yoga - I | 1 |
| I | UC | Social Leadership Development Program | 1 |
| I | UC | Financial Literacy | 1 |
| I | PF | C Programming | 5 |
| I | PF | Computer Organization | 3 |
| I | PF | Digital Electronics | 4 |
| I | PF | Discrete Mathematics | 3 |
| I | PF | Applied Statistics | 2 |
| | | Total Credits: | 22 |

Semester II

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|-----------------------------------|---------------|
| II | UC | Yoga - II | 1 |
| II | UC | Co-creation | 1 |
| II | UC | AI for everyone | 2 |
| II | UC | Foundation of Peace | 2 |
| II | UC | Indian Knowledge System (General) | 2 |
| II | UC | Sports | 1 |
| II | PF | Database Management system | 4 |
| II | PM | Microprocessor | 4 |
| II | PF | Data Structure using C | 4 |
| | | Total Credits: | 21 |

Semester - III

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| III | UC | Spiritual and Cultural heritage: Indian Experience | 2 |
| III | UC | Research Innovation Design Entrepreneurship (RIDE) | 1 |
| III | UE | University Electives - I | 3 |
| III | PM | Microcontroller | 4 |
| III | PM | OOP using C++ | 4 |
| III | PF | Web Technologies | 4 |
| III | PF | Data Analytics | 2 |
| III | PF | Lab on Python | 2 |
| | | Total Credits: | 22 |

SEMESTER - IV

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|-------------|--|---------------|
| IV | UC | Rural Immersion | 1 |
| IV | UC | Life Transformation Skills | 1 |
| IV | UE | University Electives - II | 3 |
| IV | PF | Indian Knowledge System (Sci. & Tech.) | 2 |
| IV | PF | Number Theory & Calculus | 3 |
| IV | PM | Data Communication and Networking | 4 |
| IV | PF | Operating System | 3 |
| IV | PM | Network Security | 2 |
| IV | PM | Software Engineering | 3 |
| | | Total Credits: | 22 |

Semester - V

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|---|---------------|
| V | UC | Managing Conflicts Peacefully: Tools and Techniques | 2 |
| V | UE | University Electives - III | 3 |
| V | PE | Program Elective - I | 4 |
| V | PM | Core Java | 4 |
| V | PM | Advanced Web Technologies | 3 |
| V | Program Capstone Project/Seminar and Internships | Cognitive Skills | 1 |
| V | Program Capstone Project/Seminar and Internships | Mini Project-I | 4 |
| | | Total Credits: | 21 |

Semester - VI

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|--|---------------|
| VI | UC | National Academic Immersion Program | 2 |
| VI | PE | Program Elective - II | 4 |
| VI | PM | Advanced Java | 5 |
| VI | PM | Artificial Intelligence and Machine Learning | 3 |
| VI | PM | Mobile Application Development | 3 |
| VI | Program Capstone Project/Seminar and Internships | Mini Project-II | 5 |
| | | Total Credits: | 22 |

Semester - VII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|------------------------------|---------------|
| VII | PE | Program Elective - III | 4 |
| VII | PM | Agile Methodology | 2 |
| VII | PM | Internet of Things | 3 |
| VII | PM | Theoretical Computer Science | 2 |
| VII | PM | Research Methodology | 2 |
| VII | Program Capstone Project/Seminar and Internships | Research Paper Writing | 3 |
| | | Total Credits: | 22 |

Semester - VIII

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|--|----------------------------|---------------|
| VIII | PE | Program Elective - IV | 4 |
| VIII | Program Capstone Project/Seminar and Internships | Internship | 15 |
| | | Total Credits: | 19 |

Program Elective Tracks

| Semester | Course Type | Course Name / Course Title | Total Credits |
|----------|------------------------|-------------------------------------|---------------|
| V | Program Elective - I | Introduction to Cyber Security | 4 |
| V | Program Elective - I | Introduction to Data Science | 4 |
| V | Program Elective - I | Introduction to Quantum Computing | 4 |
| VI | Program Elective - II | Cyber Law & IT Act | 4 |
| VI | Program Elective - II | Data Science with Open-Source Tools | 4 |
| VI | Program Elective - II | E-commerce and E-business | 4 |
| VII | Program Elective - III | Intellectual Property Rights | 4 |
| VII | Program Elective - III | Natural Language Processing | 4 |
| VII | Program Elective - III | Intelligent IoT Systems | 4 |
| VIII | Program Elective - IV | Cyber Forensics | 4 |
| VIII | Program Elective - IV | Big Data Technologies | 4 |
| VIII | Program Elective - IV | Entrepreneurship in IT | 4 |

*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.